1.15 Solution: Take the functional given by Eq. (1.67) and generalize to multiple surfaces scenario, we have

$$I[\psi, \{f\}] = \frac{1}{2}\int_V \nabla\psi\cdot\nabla\psi d^3x - \oint_V g\psi d^3x - \sum_{i=1}^{N}\int_{S_i} f_i\psi da_i,$$

subject to the constraints

$$\oint_{S_i} f_i da_i = Q_i, \quad \text{for } i = 1, \cdots, N.$$

To find the minimum of the functional under the constraints, we can introduce Lagrange multipliers λ_i for every surface, and the corresponding Lagrangian is

$$\mathcal{L}[\psi, \{f\}, \{\lambda\}] = \frac{1}{2}\int_V \nabla\psi\cdot\nabla\psi d^3x - \oint_V g\psi d^3x - \sum_{i=1}^{N}\oint_{S_i} f_i\psi da_i + \sum_{i=1}^{N}\lambda_i\left(\oint_{S_i} f_i da_i - Q_i\right).$$

The variation of the Lagrangian is given by Eq. (1.68), with extra terms,

$$\begin{aligned}\delta\mathcal{L}[\psi, \{f\}, \{\lambda\}] &= \int_V \left[-\nabla^2\psi - g\right]\delta\psi d^3x + \sum_{i=1}^{N}\oint_{S_i}\left(\frac{\partial\psi}{\partial n} - f_i\right)\delta\psi da_i \\ &\quad - \sum_{i=1}^{N}\oint_{S_i}\delta f_i(\psi - \lambda_i)da_i + \sum_{i=1}^{N}\delta\lambda_i\left(\oint_{S_i} f_i da_i - Q_i\right).\end{aligned}$$

For $\delta\mathcal{L}$ to vanish, we need to have

$$\nabla^2\psi = -g \text{ within } V, \quad \frac{\partial\psi}{\partial n} = f_i \quad \text{and} \quad \psi = \lambda_i \text{ on } S_i.$$

Therefore, we have established that every surface is equipotential.